

Who Owns the Future of Work?

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Figure 1: "The Future of Work" by Nicolle R. Fuller, National Science Foundation. [10]

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ABSTRACT

This submission is intended to start a critical conversation about the "Future of Work." I expose the ideological commitments of major funders of research into this area, such as the NSF and private think tanks, because they shape narratives on work and worker organizing as well as the course of technology development more broadly. In the research itself, workers may be invited to participate in computing research, but they seldom determine the goals of projects

nor do they own the results. Ultimately, while government agencies and nonprofit organizations claim to seek improved conditions for workers, the knowledge and technologies produced through this research serve primarily to undermine the power of working class people and their unions.

CCS CONCEPTS

• **Social and professional topics** → **Computing / technology policy; Employment issues; Economic impact.**

KEYWORDS

future of work, unions, labor organizing

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1 INTRODUCTION

HCI researchers are no stranger to political topics, including questioning the funding sources of our research [12]. As more and more of us begin to explore “the future of work”—including the recent CHIWORK initiative [17]—it is extremely important that we understand the social and political forces at play. I examine two funders to accomplish this, from the public and private sector respectively: the United States’ National Science Foundation (NSF) and the WorkRise Institute. Importantly, my goal with this provocation is not to dissuade researchers from seeking funding from these organizations, nor am I arguing that those who do take money from them are somehow morally bankrupt. I have no time for a moralizing sense of “ethics” that is tied to the decisions of individual researchers, or to the merits and demerits of individual research projects. I want us to think about bigger questions: when it comes to research on the “future of work,” what is the ideological water we swim in? Where are the currents taking us?

2 THE NATIONAL SCIENCE FOUNDATION

From 2020 to 2021, at the time of this writing, the NSF has awarded 114,893,819 USD to researchers across the country under their program “Future of Work at the Human-Technology Frontier” [9]. Among its stated aims are to “understand and develop the human-technology partnership,” “design new technologies to augment human performance,” and “understand and influence the impact of artificial intelligence on workers and work.” In the private sector as well, think tanks like the WorkRise Institute explore “bold ideas for transforming the labor market” with millions of dollars of funding from corporate backers, including the Walmart Foundation, the MasterCard Center for Inclusive Growth, and General Motors [19]. Far from a conspiracy theory, journalists and media critics have shown how corporations use both philanthropic initiatives and research to advance their interests [24]. In the latter, researchers are enlisted to lend scientific, “objective” authority to anti-worker policy decisions and technology development.

As one of the largest funders of scientific research in the country, the NSF wields an incredible amount of power in determining the topics and scope of projects. Funding calls, like the Future of Work

program described above, solicit proposals in particular areas of national interest. If they are to keep their jobs, researchers must find the funding opportunities that best fit them, or else pitch their work in a way that fits the opportunity. Sociologist Scott Cummings has recognized the “research enterprise as part of larger business and commercial structures,” and as fundamentally “commercial and entrepreneurial” in nature [7]. But this goes beyond researchers individually selling their work to funders to stay afloat. Cummings argues that it extends to the myriad partnerships that universities hold with corporate and government partners, the erosion of working conditions for academic workers, and the prioritization and expansion of academic departments that align with private sector interests (i.e., business, science, and engineering) at the expense of the arts and humanities. Cummings’ 1984 article was originally a response to the Reagan Administration’s attacks on state-supported research; the NSF, National Institutes of Health, the National Endowment for the Humanities, and many others were all on the chopping block.

John Ashbrook of Ohio, then one of Reagan’s lackeys in the House of Representatives, decried in 1980 what he saw as the NSF’s frivolous expenditures, including research on the working conditions of married women (“an unwarranted initiative into the American home”) and on couple formation among gay people (“just plain stupid”) [2]. His preferred solution was to dissolve funding for the NSF altogether and replace it with the likes of “professional” conservative think tanks such as the Brookings Institution and Heritage Foundation, which are “moving knowledge in the humanities forward with and without federal subsidy.” Attacks on publicly-funded research from the right are ongoing, especially on matters of so-called “controversial public policy.” Ashbrook characterizes publicly-funded research as way to force liberal ideology¹ onto Americans while elevating private think tanks as a wholesome, righteous alternative that push the correct ideas forward without government meddling. Despite these unwarranted attacks, however, I believe that it is possible and necessary to critique the NSF from the left; and, as I will show, anti-worker research on the “future of work”—currently supported by both the NSF and private foundations—has broad consensus and support from both liberals and conservatives.

The banner image stretching across the NSF’s “Future of Work” funding opportunity page is a techno-utopian “smart city,” created for the foundation by scientific illustrator Nicolle Fuller (see Figure 1). In the image, lush green fields share space with schools offering each child a personal interactive bubble, which captures their thoughts as they gather around a holographic aquarium; data analysts interact with a massive interface bedazzled with data visualizations; and, in the shadow of a bio-dome, a man and a child kneel in a hexagonal field of crops as a drone spits out projections of charts and graphs. The workers harmoniously interact with the technologies around them, and each of those systems in turn harmoniously interact with each other. It’s an image of the big-data-driven city of the future, with a multiracial collection of workers—municipal, agricultural, and educational—at its heart.

¹As a case in point: in 2018, I received an email from the conservative outlet *Washington Free Beacon* about my NIH-funded grant to explore technologies to support transgender people. The reporter wrote, “Can you provide comment on why you think this work is important and worthy of federal funding?”

Interestingly, despite the NSF's goal to "position our Nation at the cutting edge of global science" [9], this city of the future is noticeably bare of national signifiers, such as flags or a recognizable landmark from any major American city. This could be anywhere, but it's also not: there is something specifically settler-colonial and American about using the prospect and promise of prosperity beyond a "frontier" to drive a narrative of national progress. Scholars of geography have understood the frontier not just as a metaphor or symbol of past colonial violence, but as an "ongoing feature of racial governance and capital accumulation over time" [18]. To put it differently: my intention is not to call out the NSF's use of the term frontier—or specific details within the banner image itself—as problematic. I want to call attention to how the history of the United States, and its ongoing colonization of Indigenous lands and genocide of Indigenous peoples, are erased by the American state's vision of a techno-utopia. It presumes that a harmonious future can be brought forth without actual reparations (e.g., returning land and power back to the Indigenous peoples from whom it was stolen), and instead by securing the future prosperity/dominance of the American state and its economy².

What does this have to do with the "Future of Work"? We need to understand concretely how the American state utilizes research about labor and technology to advance its interests. In a video showcasing the Future of Work program as one of "Ten Big Ideas for Future NSF Investment," Fay Cook, NSF's Assistant Director for Social, Behavioral, and Economic Sciences, proclaims:

"We are on the cusp of a major transformation in the workplace. The workplace of tomorrow is going to feature a symbiotic co-existence of humans and machines and cyberspace... Today, we have a unique opportunity to actively shape that future to improve the quality of work, and ultimately, to increase productivity and economic growth." [8]

Cook gives away the game in her last sentence. Evidently, the value of research proposals submitted to the Future of Work program will ultimately be judged by whether and how they perpetuate (and strengthen) already existing economic relations. The endless growth of the settler state and its technologies are never questioned, paving the way for the continued devastation of the climate crisis. Meanwhile, Indigenous water protectors and their allies have repeatedly put their bodies on the line to halt capital accumulation, only to be greeted by militarized police [22].

But even within this paradigm of settler-colonial fantasy, is it possible that the technologies generated through millions of dollars worth of publicly-funded research will provide some benefit to workers? The answer is a resounding no. Funded research projects under the NSF's Future of Work program often aim to create technologies that surveil and track workers' activities, and to increase their job duties (and thus provide justification for reduced staffing rates). Furthermore, projects are framed in a way that advances specific technology "solutions" prior to bringing ideas to workers (if workers are consulted at all). For example, a 3 million USD grant intends to "increase farm workers' trust" in data-driven tools in the

"living laboratory" of currently operating farms. The researchers argue that "data-based agricultural technologies"—not dissimilar from those depicted in the futuristic image above—can "generate site-specific farm recommendations that can improve farm productivity and environmental sustainability."

The best that workers can hope for with these projects is to be offered a chance to provide minimal "input" toward the direction of the technology, a research strategy that is often touted as "participatory" (as I and others have written about elsewhere [1, 13]). But workers are unable to contest their foundational assumptions and end-goals: improving productivity. That, as Cook explicitly said, is the primary goal of efforts to shape the "future of work."

3 WORKRISE'S "BOLD IDEAS"

While the NSF frames its funding opportunities as gestures toward a techno-utopian future, think tanks and nonprofits like the Urban Institute (and its WorkRise initiative) seem to be concerned with more immediate problems: income inequality, lack of opportunities for upward mobility, and low wages for Black workers and workers of color. WorkRise frames its mission as directly advancing workers' interests, but a close reading of their website, major donors, board members, and funded projects suggests otherwise. The contradictions between WorkRise's stated pro-worker values and its corporate donors is resolved through a strategy of governmental and corporate benevolence, rather than workers organizing against the corporate class. In this ideological framework, moreover, working toward a prosperous "future of work" means returning to a prosperous past.

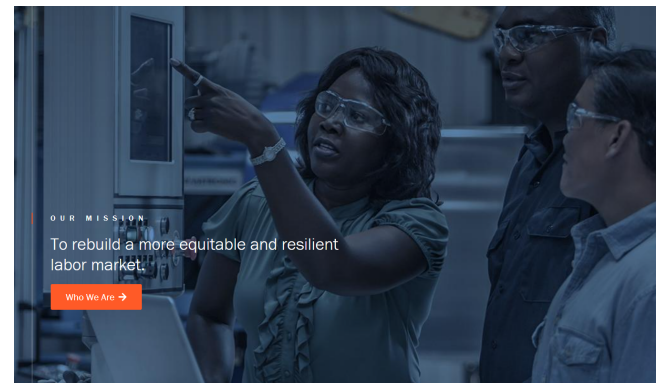


Figure 2: The banner image on WorkRise's website.

Visitors to WorkRise's website are greeted with a large banner image, with the overlaid text: "Our Mission: To rebuild a more equitable and resilient labor market" (see Figure 2). The word "rebuild" suggests that an equitable labor market previously existed; this contention is elaborated by WorkRise board member David Rolf, founder and president emeritus of SEIU (Service Employees International Union) Local 775. In a WorkRise interview—published under the category "Stakeholder Voices"—Rolf says:

"During the decades immediately following the Second World War, we saw unrivaled economic growth in the country, and the distributional outcomes of

²The foundation's funding calls are sometimes more overtly militaristic: their "Strengthening American Infrastructure" program aims to, among other things, "enhance national security."

that growth were approximately equal across income groups. We then turned the whole thing into reverse, following the wrong leaders and wrong theories and passing the wrong laws and policies. After a failed resistance to the New Deal, economic conservatives successfully implemented their ideas beginning in the 1970s. This is what led to an era of deregulation, privatization, and a decline in unionization. It also gave rise to the idea that shareholder value supremacy in corporate governance was the best way to produce broadly shared prosperity.” [14]

As Rolf would have it, the postwar labor landscape was one of bountiful and equitable growth and prosperity. While it's true that wages grew in tandem with economic growth in the postwar period, the truth is a bit more complicated. After the war, as labor historian Jeremy Brecher writes, overall wages were in fact decreasing: “Through loss of overtime and downgrading of workers, the weekly wages of non-war workers decreased 10 percent between spring 1945 and winter 1946; war workers lost 31 percent and were making 11 percent less spendable income than they were in 1941” [6]. Despite (and also because of) the great gains derived from the labor movement's strength at that time, workers mobilized to defend what they won from new corporate attacks: 2,970,000 workers went on strike just in the first six months of 1946, and a total of 4.6 million struck by the year's end. Such a display of power would be met with the Truman Administration's forceful seizure of several industries, as well as threats to call in the Army and forcibly draft strikers unless they went back to work.

Millions of workers were striking, not just for wages, but for power and control over working conditions. As historian Toni Gilpin describes in her book *The Long Deep Grudge*, the Farm Equipment Workers union (FE) went on strike against their employer, International Harvester, in January 1946 [11]. They demanded “democratic control over the powers of management,” a “fair, understandable piecework system,” and “straight plant-wide seniority.” Since at least 1916, the FE had been fighting the piecework system, first introduced in the late 1800's. In piecework, bosses offered each worker a different and fluctuating pay rate for each functional part they produced. In effect, bosses could dynamically optimize their profits—and how hard and how much each worker had to work for equivalent wages—every day. At stake here was more than pay: if piecework was changed in workers' favor, it would “[strike] at the heart of managerial discretion, making it impossible... for the company to maintain unilateral control over the execution of work.” And the demand for plant-wide seniority, Gilpin explains, was especially crucial to the plant's Black workers, who were passed up for promotion in favor of white workers in other divisions even though they'd been working at Harvester longer.

The FE, a Communist-influenced union with high levels of Black leadership, was frequently assailed both by the government and other unions. In the decades after the war, Gilpin shows, anti-communist campaigns led by the United Auto Workers (UAW) union attempted unsuccessfully to wrest control from the FE in its midwest plants. Communists (and before them, anarchists) were essential to labor's victories around the turn of the 20th century, but many mainstream unions were quick to jettison any perceived

connection with labor's left wing. Then-UAW President and rabid anti-communist Walter Reuther said that “under the rule of law, competing classes could work for the common good” and that he had “nothing against free enterprise.” Essential to his view was that the American economy must continue to grow, so that the government could provide expanded benefits to workers.

These rifts persist in today's labor movement, and David Rolf is Reuther's ideological successor: a “labor leader” who believes that partnerships with business will lead to improved conditions for working people. In early 2018, Rolf penned a letter with Dara Khosrowshahi, the CEO of Uber, and self-described venture capitalist Nick Hanauer [20]. They began with the eternal refrain: “The world of work is changing.” Their solution? A system of “portable benefits” that would allow Uber drivers to access healthcare and workers compensation—all without a union! The letter deftly undermines gig economy workers' ongoing struggle for union representation, first through their fight for the passage of California's AB5 law, which would have rightfully reclassified many “independent contractors” as employees, and then against Proposition 22, which exempted companies like Uber, Lyft, and DoorDash from adhering to AB5. Prop. 22 passed with 205 million USD in backing—the most ever spent on a California ballot question—to convince voters in any way possible to vote in favor. The companies' underhanded tactics included using their app to push propaganda directly to drivers' phones, and mailing out “Yes on 22” flyers en masse, signed by fake yet progressive-sounding organizations (such as “Feel the Bern, Progressive Voter Guide”) [16].

Ultimately, Prop. 22's passing is a proof-of-concept for similar legislation across the country [5]. The fundamental premise, as the letter lays out, is that “the American social safety system... has not kept pace” with contemporary working conditions and technology. Rolf and his capitalist pals continue:

“These hurdles will only be overcome when parties are willing to sit down, put aside historical differences, and work together to develop a solution... The pursuit of local solutions will expedite the move from the theoretical into the practice, unraveling the thorny issues and beginning to show how a portable benefits system can empower workers and enable technology to meet the growing demand for more flexible, independent forms of work.” [20]

This passage makes a number of assumptions here that are worth breaking down. For one, it locates positive social change in collaboration (“when parties are willing to sit down”) rather than struggle between workers and bosses. Those pesky “historical differences” (such as when millions of workers went on strike) need to be “put aside.” Secondly, their emphasis on “local solutions” is a not-so-subtle plug in favor of state-level initiatives (like Prop. 22 and its ilk) that would deliver from above moderate, “portable” benefits to gig workers as they move from app to app. They argue that this system would “empower workers”; of course, workers organizing from below to build and flex their own power to win changes that they want, on their own terms, is off the table.

Finally, Rolf et al. write that the supposed empowerment for workers would go hand in hand with “[enabling] technology to meet the growing demand for more flexible, independent forms of work.”

They retroactively justify the existence of Uber and other companies that promise “independent” work. But as Shona Clarkson, organizer for Gig Workers Rising, said:

“We know that drivers do not actually have independence while driving for Uber. There is no independence in working 70+ hours a week, not being able to set your own rates, not being able to see where a ride is going and having *no real control at work*. The benefits promised under Prop 22 were a sham that have not materialized” (emphasis added). [5]

It all comes down to control. Rolf and Khosrowshahi—just like Walter Reuther and the wealthy industrialists of his day—would allow workers none of it. It now becomes more clear why Rolf shares his WorkRise board seat with the former Ford Motor Company VP Bill Dirksen and Michael R. Strain, an economist from the conservative think tank American Enterprise Institute. And it’s now obvious why WorkRise is funded by the likes of MasterCard, the Walmart Foundation, and General Motors: these corporations benefit from the ideologies peddled and perpetuated through the scholars and researchers they fund.

In study after study, corporate-backed researchers collect endless amounts of data about what benefits workers should receive and how they should receive them. Through networks of lobbyists and purchased legislation like Prop. 22, corporations are putting this research (and the ideology driving it) into practice, tightening their grip over workers’ everyday lives. Most frustratingly, both liberals and conservatives frame this as a victory for workers, as “empowerment.”

In 2020, for example, WorkRise funded a project to conduct computer “microsimulations” of parents receiving child care subsidies. The researchers are creating a data-driven “forecast” of the future “to model potential policy scenarios.” They suggest that the “findings will be critical to understanding the geographic, racial, and ethnic gaps” in service provision to “inform policymaking.” But the ultimate result of such research is that workers cannot collectively determine the future for ourselves. Instead, we are left to rely on the top-down provision of benefits delivered by “policymakers”—and that is by design.

The narrative of social progress championed by Rolf and Reuther (in which management, labor, and government come together to solve the problems of the day) is not backed by history, but by ideology. This ideology informs the concerted, ongoing, decades-long effort among so-called “labor leaders,” conservatives, and capitalists to erode worker power and control in the workplace.

But beyond that, the institutions profiled here both share a particular understanding of the past and future. Their “future of work” includes visions of a harmonious techno-utopia (NSF) or a return to an earlier, more prosperous state of affairs (WorkRise). In both, technology is figured as an inevitable force of nature that requires novel adaptations on the part of workers and governments. However, in reality, the owners of technology make deliberate decisions—and influence others through political machinations like Prop. 22—that alter workers’ lives every day. In this, we can hear echoes of International Harvester’s insistence in 1946 to retain the piecework

system and the control over workers it afforded them³. These power imbalances, these realities of class struggle, are flattened when solutions are presented as collaborations between “humans” (NSF) or “stakeholders” (WorkRise).

Moreover, these harmonious “collaborations” are meant to take place without any actual change in the structure of class society, rooted in slavery, genocide, and the dispossession of indigenous lands. As such, both future-visions are also propped up by nationalism, with the United States’ productivity and economic growth paramount. As David Rolf made clear, the glory of the nation’s past is constructed around its victory in World War II and the “prosperous” decades immediately afterward. Of course, there is no mention of the strikes that brought unprecedented numbers of workers to the streets (nor does he figure in the coups fomented around the world in service of American hegemony). American workers, it seems, are the only workers of the future, as only their benefits and wellbeing are considered. In effect and by design, this forces them into a position of begging for scraps, and into more uphill battles if they attempt any internationalist solidarity across global supply chains.

Unopposed, the United States government, think tanks, and their corporate backers would usher in a future where technology sees no change in ownership, guaranteeing its continued use as a tool for subjugation, surveillance, and ecological destruction. The only way forward is for the future of work—and the future writ large—to be owned by the workers of the world.

4 IMPLICATIONS FOR DESIGN

Just kidding—there are none. There are no ways that we, as computing researchers and practitioners, can individually modify our professional practices or research methodologies that would accelerate the end of centuries of racist and anti-worker policy. In other words, what we have to do won’t get you promoted. In fact, it might more likely cause your university administration to dislike you very much.

Ask: how have student workers, dining hall, and custodial staff in your institution organized in the face of the COVID-19 pandemic? How many faculty, contingent and adjunct especially, were forced to return to campus and teach in person as variant cases spiked? How many graduate student workers did university administrators threaten to replace[21] or intimidate with armed police[15] for the apparently egregious act of going on strike to demand better healthcare and basic protections from harassment?

As academic workers, we must not consider the “future of work” as an abstract force of nature, as a process by which machines spontaneously spring up out of the ground to replace factory workers and automate their jobs but sparing those of us secured in the ivory tower. These are real decisions being made, by real people with real motives, that affect us all. To combat them, it is incumbent upon us to build and strengthen the labor movement, now, today.

Here are some potentially helpful resources:

- The Emergency Workplace Organizing Committee: <https://workerorganizing.org/>

³In 2021, nearly 80 years later, we saw this ongoing fight flare up at farm equipment factories in the American Midwest, owned by the John Deere corporation. Workers organized with the United Auto Workers (UAW), went on strike for five weeks over piecework, health care, and other issues.[3]

- Labor Notes, Secrets of a Successful Organizer: <https://labornotes.org/secrets>
- The many researchers in the HCI and CSCW communities already taking up this call [4, 23]
- Your fellow workers, from whom you can learn about the issues they face, and how you might build a better future with them.

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